CHAPTER B

# Fluency bank

Two questions per session, in 6 to 8 minutes including correction. The identifiers allow items to be revisited; they are not a list to be completed in full in every lesson. The availability dates are based on the planned teaching; postpone them if the topic has not yet been studied.

### R001 · Fractions

Calculate $\frac{2}{3} + \frac{1}{4}$.

### R002 · Order of operations

Calculate $5 - 2(3 - 7)$.

### R003 · Powers

Simplify $\dfrac{10^5 \times 10^{-3}}{10^4}$.

### R004 · Roots

Simplify $\sqrt{50}$.

### R005 · Proportion

18 represents 30% of a total. What total?

### R006 · Direct rate

A price of 80 falls by 15 %. Final value?

### R007 · Initial value

After an increase of 20%, the value is 72. Initial value?

### R008 · Calculated rate

A quantity goes from 40 to 50. Rate?

### R009 · Successive rates

20% increase then 20% decrease: overall rate?

### R010 · Reverse rate

What rate offsets a 20% decrease?

### R011 · Zero product

Solve (2 $x-1)(x+3)=0$.

### R012 · Expanding

Expand (2 $x-3)^2$.

### R013 · Factorising

Factorise 3 $x^2-12x$.

#### R014 · Sign of an affine function

Solve $-3x+6\geq 0$.

### R015 · Gradient

Slope through (1; 4) and (3; 0)?

### R016 · Line

Equation with slope -2 through (1; 4)?

### R017 · Weighted average

The values 2 and 5 have frequencies 3 and 2. Mean?

### R018 · Median

Median of 1, 2, 2, 5, 8, 10?

### R019 · Quartiles

For the data set 1, 2, 3, 4, 5, 6, 7, 8, give $Q_1$ and $Q_3$ with the ceiling indices $\left\lceil \frac{n}{4} \right\rceil$ and $\left\lceil \frac{3n}{4} \right\rceil$.

### R020 · Statistical standard deviation

For 1 and 3 equally weighted: mean, variance and standard deviation?

### R021 · Units

Convert 1 min 40 s into seconds; 2400 frames over this duration give how many per second?

### R022 · Simple probability

With a fair die, what is the probability of a multiple of 3?

### R023 · Complement

$P(A) = \frac{3}{8}$. Give $P(\overline{A})$.

### R024 · Interval

Write $x \geq -2$ and $x < 3$ in interval notation.

### R025 · Logic

Negate "all the pupils passed".

### R026 · Histogram

Two classes with widths 2 and 4 have the same frequency, 20. In a histogram with areas proportional to the frequencies, what are the relative heights?

### R027 · Pie chart

A sector measures 72 °. Proportion?

### R028 · Scale

An axis starts at 90. Bars end at 100 and 110. Is the second value double the first?

### R029 · Percentage of a percentage

40% of pupils do A; among them, 25% do B. What proportion do A and B?

### R030 · Proportion and frequency

One category represents 0.125 of 80 people. Frequency?

### R031 · Defining and representing a sequence

Calculate the first four terms of $w_0 = 2,\ w_{n+1} = w_n + 2n + 1$; suggest an explicit formula.

### R032 · Arithmetic sequences

We know $u_3 = 17$ and $u_8 = 37$ for an arithmetic sequence. Determine $r$ and $u_0$.

### R033 · Geometric sequences

Study the variation of $u_n = -2 \times 3^n$, then of $v_n = 2(-1)^n$.

### R034 · Sums, subscripts and proofs

Calculate $\sum_{k=2}^{5} 2^k$ without a summation symbol in the answer.

### R035 · Conjecturing a limit and programming a threshold

For $u_n = 48(\frac{1}{2})^n$, determine the first indices where $u_n \leq 6$ and where $u_n < 6$.

### R036 · Three forms, three readings

Give three forms of $g(x) = x^2 + 2x - 3$. Read off its roots and its minimum.

### R037 · Discriminant and solving

Solve 3 $x^2 - 6x + 3 = 0$ and $x^2 + 2x + 5 = 0$.

### R038 · Factoring without automating the discriminant

Determine the quadratic polynomial f such that $f(1) = f(4) = 0$ and $f(0) = -8$.

### R039 · Sign, inequalities and optimisation

Solve $x^2 - 4x - 5 < 0$, then maximise $x(8 - x)$ on [0; 3].

### R040 · Projection and angle

Two vectors have norms 2 and 5 and form an angle of 90 °, then 180 °. Calculate the dot products.

### R041 · Coordinates, norm and orthogonality

For which real t are the vectors ( $t$; 2) and (3; $-6$) orthogonal?

### R042 · Vector calculation and identities

We are given $\|\vec{u}\| = 2$, $\|\vec{v}\| = 3$, $\vec{u} \cdot \vec{v} = 1$. Calculate $\|\vec{u} - \vec{v}\|$.

### R043 · Al-Kashi and choice of method

A triangle has $AB = AC = 4$ and $BC = 4\sqrt{2}$. Determine angle A.

### R044 · The circle with diameter AB

For $A(0; 0)$, $B(6; 0)$, find the points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 7$.

### R045 · Rate of change and secants

Compute the rate of $f(x) = x^2 + 3x$ between 2 and $2 + h$.

### R046 · Derivative at a point

For $f(x) = 3x^2 - 2x$, calculate $f'(1)$ from the rate.

### R047 · Tangent and affine approximation

We are given $f(3) = 5$ and $f'(3) = -2$. Write the tangent and approximate $f(3{,}04)$.

### R048 · When differentiability fails

Justify that $g(x) = |x - 2|$ is not differentiable at 2. Also study the differentiability of $x \mapsto \sqrt{x}$ at 0 via its rate.

### R049 · Conditioning: changing the denominator

Out of 60 objects, 20 are red; of the red ones, 5 are damaged. Calculate $P(R \cap D)$ and $P_R(D)$.

### R050 · Trees, partitions and total probabilities

$P(A) = \dfrac{1}{3}$, $P_A(B) = \dfrac{3}{4}$, $P_{\overline{A}}(B) = \dfrac{1}{4}$. Calculate $P(B)$.

### R051 · Independence and Incompatibility

We are given $P(A) = 0{,}4$, $P(B) = 0{,}5$, $P(A \cap B) = 0{,}18$. Are they independent?

### R052 · Repeated trials and Bernoulli

Four independent trials with probability $p = \dfrac{1}{3}$. Calculate the probability of four failures, then of at least one success.

### R053 · Simulate and discuss the model

A simulation of 2000 points gives 675 points under $y = x^2$. What area does it estimate?

### R054 · Derivative function and standard functions

Diff. $x^5$, $x^{-2}$ & $\sqrt{x}$ stating the domains.

### R055 · Sum, product, reciprocal and quotient

Differentiate $(x^2 + 1)(3x - 2)$ then $\dfrac{1}{x^2 + 1}$.

### R056 · Sign of the derivative and extrema

Study $f(x) = x^2 - 4x + 1$ on [0; 3] and give its extrema over this interval.

### R057 · Parity, inequalities and comparison

Study the parity of $x^4 + x^2$ & $x^3 - x$ ; prove $x^2 + 1 \geq 2x$.

### R058 · Optimisation and Newton's method

A rectangle with perimeter 12 m has width x. Find the maximum area using differentiation.

### R059 · Normal vector and Cartesian equation

Equation of the line through (2; −1) with normal vector (3; 4).

### R060 · Orthogonal projection and distance

Project $P(4; 1)$ onto $d : x = 2$.

### R061 · Equation of a circle

Recognize $x^2 + y^2 + 2x - 8y + 17 = 0$.

### R062 · Solve a configuration

Intersect ( $x - 1)^2 + (y + 2)^2 = 9$ with $y = 1$.

### R063 · Variable, event and distribution

Two fair coins are tossed. X counts the tails. Write its distribution and $P(X \leq 1)$.

### R064 · Expectation and linearity

A gross gain is 10 with probability $\frac{1}{5}$ and 0 otherwise. What stake makes the game fair?

### R065 · Variance, standard deviation and König-Huygens

For X with values $-1, 1$ equally likely, calculate the expectation, variance and standard deviation.

### R066 · Transforming and comparing winnings

$E(X) = 3$, $V(X) = 4$. Give the parameters of $Y = -2X + 5$.

### R067 · Samples and observed mean

For $\mu = 5$, $\sigma = 3$ and n=36, give the requested experimental interval.

### R068 · Defining a new function

What is the equation of the tangent to $e^x$ at 1?

### R069 · Algebraic properties and comparisons

Simplify $\frac{e^{3x}}{e^{x-1}}$; solve $e^{x+2} > e^{2x}$.

#### R070 · Differentiating and studying

Diff. $g(t) = 3e^{-2t}$ and study how it varies.

### R071 · From discrete to continuous

For $Q(t) = 5e^{0,3t}$, express $Q(n)$ and $\frac{Q(t+2)}{Q(t)}$.

### R072 · Building numerically and proving an inequality

Show that $e^{2x} \geq 1 + 2x$ for all real x.

### R073 · Measuring an angle in radians

Convert 45 °, $120°$ ; give the arc with radius 2 m and angle $\frac{3\pi}{4}$.

#### R074 · Wrapping and the image of a real number

Plot the images of $\frac{7\pi}{2}$ & $-\frac{5\pi}{3}$ by reducing them to one turn.

### R075 · Sine, cosine and standard values

If cos $x = \frac{3}{5}$ and the image of x is in the fourth quadrant, give sin $x$.

### R076 · Associated angles and symmetries

Calculate the two coordinates of the images of $\frac{4\pi}{3}$ and $7\frac{\pi}{4}$.

#### R077 · Approximating pi and reinvesting

For n=6 and c=1, calculate the new chord after doubling.

## R078 · Graphical reading

In Figure R078, read $f(1)$, solve $f(x) = 0$ and $f(x) > 0$.

